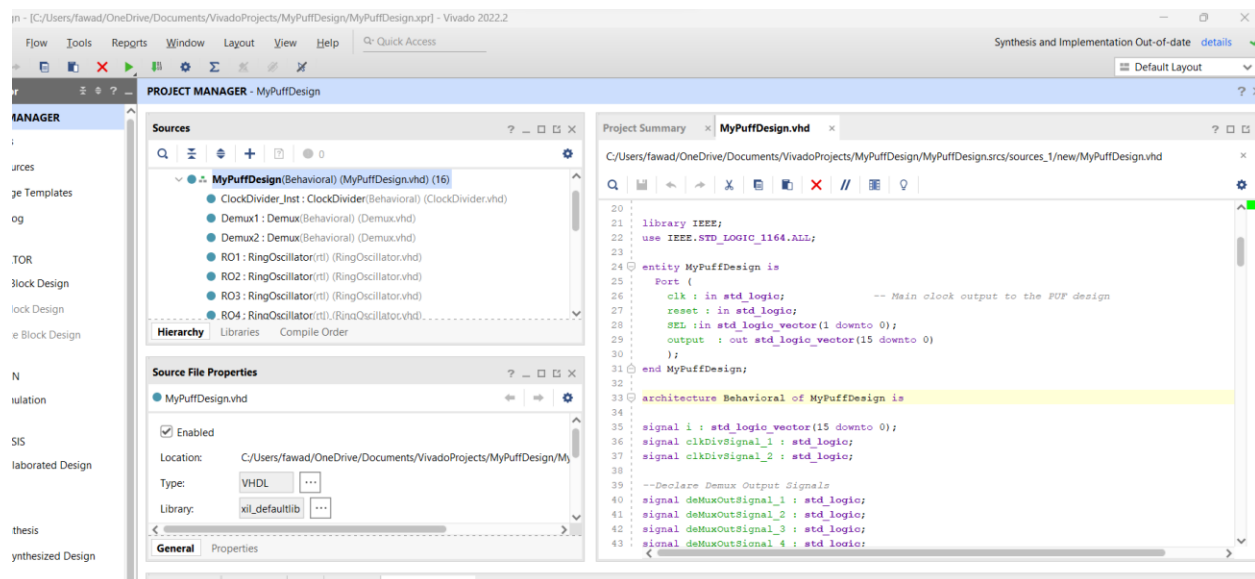


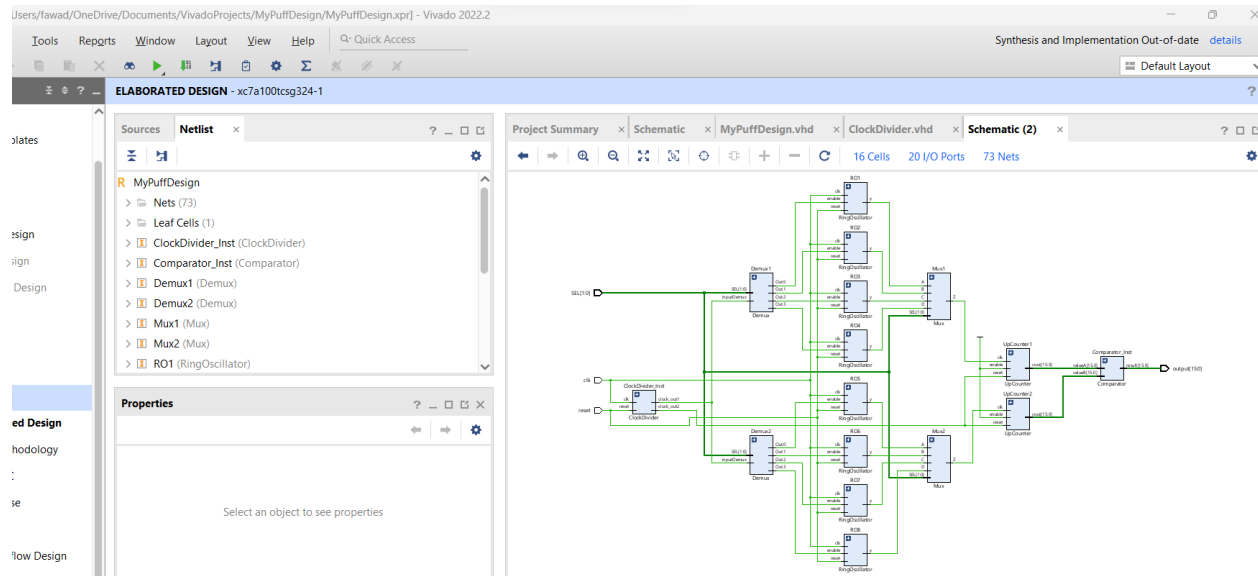
OBJECTIVE-The main purpose of this lab is to give the student an opportunity to take everything that was done in this course and bring it all together under one integrated design. This lab is the final culmination of all the concepts learned in this class about the VHDL programming of an FPGA.

PROCEDURE-In this lab, the student designs a ROPUF. This ROPUF is made up of a clock divider which splits the clock signal between a demultiplexer and a counter. The multiplexer gets its signal from Sel. There are two demultiplexers, two multiplexers, two comparators, and eight three ring oscillators. Each demultiplexer sends its output to the four ring oscillators. The four ring oscillators, the two demultiplexers, two multiplexers and two counters are created by instantiating the main code for the ring oscillator, the demultiplexer, the multiplexer, and the counter. The output from each ring oscillator goes into each one of the four multiplexer inputs. Then the multiplexer sends its output to an up counter which sends the counted digits into a comparator. The comparator performs the comparison operations between two frequencies and then outputs the result in the top-level file. At the very end, the student implements this VHDL code on to a physical FPGA.

CODE- The following VHDL code was used for designing this ROPUF. This is the main top level PUF file which integrates all the components used in the design of this ROPUF:



SCHEMATIC:



The rest of the code for the demultiplexer, multiplexer, three ring oscillators and the up counter was simply taken from the previous lab projects which required designing those components. Those projects were done separately by the student throughout this semester.

RESULTS- After running the code for the ROPUF both in simulation mode and then on the actual FPGA hardware, the correct result was finally obtained after some false starts. There was some difficulty experienced during the synthesis process after the xdc file was added to the project. But after much troubleshooting the code ran successfully.

LEARNING OUTCOME- This project taught the student how to take various components built with VHDL code and integrate them all into one package which in this case is a ROPUF. It was an excellent way to give the student valuable learning experience in not only how to build each component of an FPGA with VHDL separately, but also how to put them all together into one unit. It was a good way to end all the lab projects done for this course.